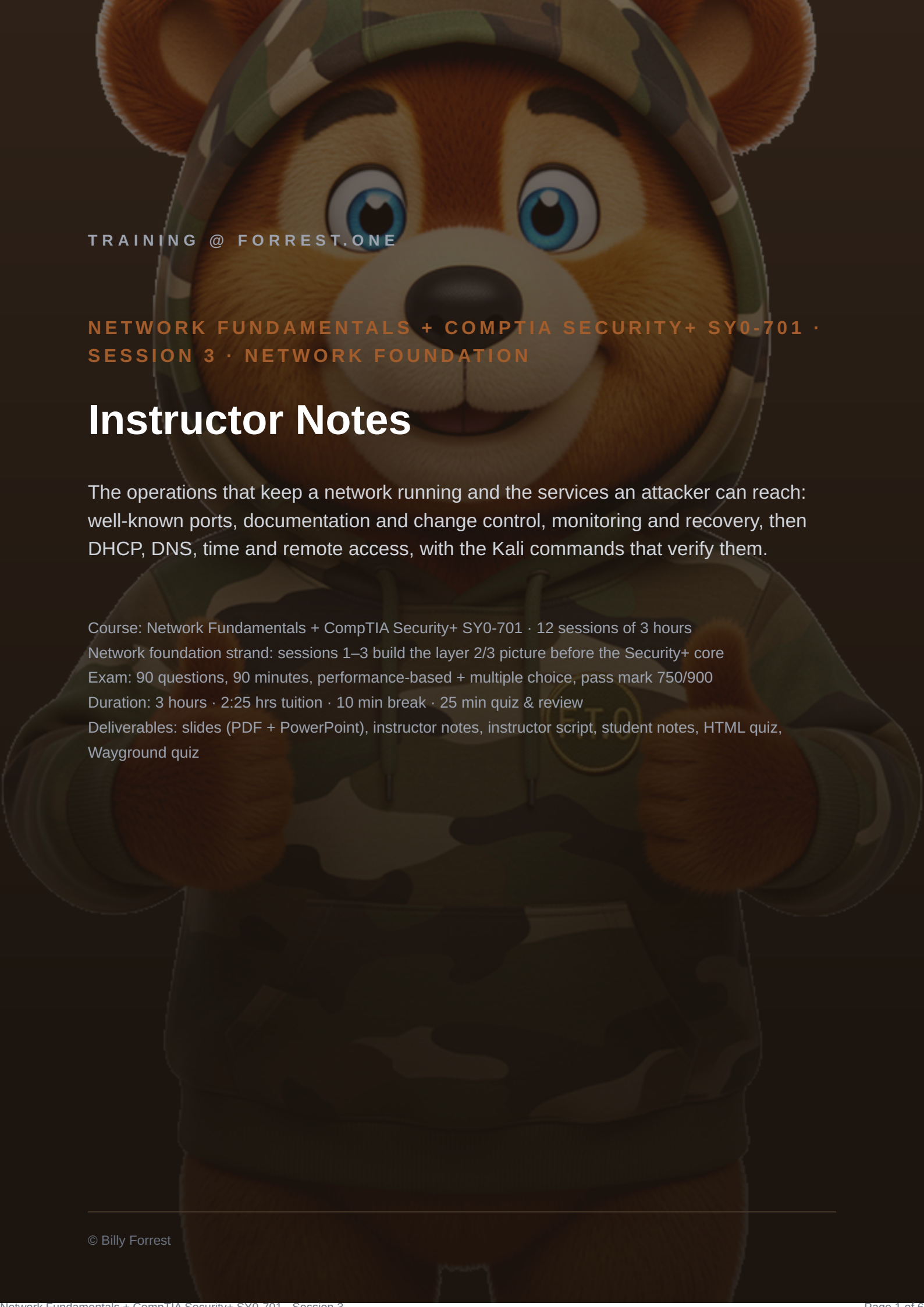




TRAINING @ FORREST.ONE

**NETWORK FUNDAMENTALS + COMPTIA SECURITY+ SY0-701 ·
SESSION 3 · NETWORK FOUNDATION**

Instructor Notes

The operations that keep a network running and the services an attacker can reach: well-known ports, documentation and change control, monitoring and recovery, then DHCP, DNS, time and remote access, with the Kali commands that verify them.

Course: Network Fundamentals + CompTIA Security+ SY0-701 · 12 sessions of 3 hours

Network foundation strand: sessions 1–3 build the layer 2/3 picture before the Security+ core

Exam: 90 questions, 90 minutes, performance-based + multiple choice, pass mark 750/900

Duration: 3 hours · 2:25 hrs tuition · 10 min break · 25 min quiz & review

Deliverables: slides (PDF + PowerPoint), instructor notes, instructor script, student notes, HTML quiz, Wayground quiz

Instructor Notes: Network Services & the Attack Surface

A 3-hour delivery guide for Session 3, closing the network-foundation strand. The session has two halves: the operations that keep a network honest (Chapters 1 to 5) and the core services themselves (Chapter 6). Run the Kali commands live if a lab is available: the commands are the practical payload of the session.

Candidates should leave able to read a port number, name the service behind it, and say what an attacker gains from it.

Session at a glance

Time	Block	Slides	Focus / activity
0:00-0:15	Welcome and objectives	1-3	Strand close; how the session is shaped
0:15-0:32	Ports and traffic types	4-7	Well-known ports, encrypted twins, traffic types
0:32-0:40	Risk discussion	8	Ranking services to close on a border device
0:40-0:58	Documentation and agreements	9-13	Diagrams, inventory, IPAM, service levels, surveys
0:58-1:13	Life-cycle and change	14-17	End of Support, disposal, change control
1:13-1:30	Configuration and monitoring	18-21	Baselines, drift, SNMP, flows against capture
1:30-1:50	Baselines, logs, resilience	22-27	Logs and SIEM, RPO and RTO, sites, HA, testing
1:50-2:00	Break	28	10-minute break
2:00-2:23	Core services and remote access	29-34	DHCP, IPv6, DNS, time, VPNs, device management
2:23-2:43	Kali verification	35-36	dig, ntpq, ss, journalctl demos
2:43-3:00	Quiz, review and close	37-40	Quiz_Session_03.html, takeaways, checklist

Materials and preparation

- Projector or screen with the Session 3 slide deck (03_Slides.pdf or 03_Slides.pptx).
- One computer or tablet per pair for Quiz_Session_03.html (any browser, no internet required).
- Printed or digital copy of the Student Notes for each candidate.
- A Kali Linux VM or machine for the live verification demos (slides 35 and 36).
- Whiteboard for the port-pair list, the DHCP DORA exchange and the availability diagrams.

Delivery guidance by slide

Slides 1-3 · Welcome (15 minutes)

Close the strand explicitly: models, addressing, switching, and now the operations and the services on top. Set the pattern that repeats all session: port, service, purpose, and what an attacker gains. Warn candidates that Chapters 2 to 5 are heavy on vocabulary, but that every term is an exam term and each one is a control or a failure mode.

Slides 4-8 · Ports, protocols and traffic types (25 minutes)

This chapter is pure memorisation with a security punchline. Teach the ports in pairs: 21 with 22, 23 with 22, 80 with 443, 389 with 636, 5060 with 5061. The pattern is the lesson: the protected form always moves to a different port. Land the rule of thumb, that if the name has no S you assume plaintext, and then use the discussion (slide 8) to make candidates rank the risk rather than recite the table. Close with the four traffic types and note that anycast is how global name servers and content networks scale, which sets up Chapter 6.

Discussion prompt (slide 8)

"Which service would you close first on an internet-facing router?" Telnet on 23 and FTP on 21 send credentials in clear text, so they go first, and both have encrypted replacements that are already standard. SNMP on 161 and LDAP on 389 are the harder judgement: they may genuinely need to be reachable, but they should be filtered to management hosts only, and SNMP version 3 should replace the community string.

Slides 9-13 · Documentation, agreements and surveys (18 minutes)

Position documentation as a security control, not paperwork. The physical diagram answers where the cable goes and the logical diagram answers how the traffic moves; without both, an incident response starts with guesswork. Cover the four specialised drawings quickly, then slow down on asset inventory and IP address management, because a failure with a serial number and an owner attached is a ticket rather than a mystery. The service level agreement slide carries one idea: the uptime figure is a design instruction disguised as a contract term. Finish with wireless site surveys as the predictive against validation distinction.

Slides 14-17 · Life-cycle and controlled change (15 minutes)

The two dates are the heart of this chapter. End of Life is a sales decision; End of Support is a security decision, because from that day every flaw found stays open. Walk the update path as a process (download,

verify the source, stage, test, deploy) and insist on a tested rollback plan. Decommissioning is a security event: a switch that still holds a certificate is a loaded gun. Then change control, and the line that most outages are self-inflicted. Slide 17 is the analyst's lens: every change touches four records, and the document you did not update is the one you will trust.

Slides 18-21 · Configuration and monitoring (17 minutes)

Drift is rarely dramatic; it is one unrecorded tweak at a time, and compliance checking is how you find it. On SNMP, make the community string memorable as a password sent in the clear, and version 3 as the answer. On monitoring, teach the division of labour: flow records give the wide shot cheaply and can be kept for months, while full packet capture reads what was actually said and can only be kept for hours. Close with the legal point that capture needs authority because packets hold personal data.

Slides 22-27 · Baselines, logs and resilience (20 minutes)

A baseline turns a graph into a judgement, so start by asking what normal looks like on your own network. Contrast fixed threshold alerting with anomaly detection, and note that a baseline is only valid until the network changes. Log aggregation and SIEM correlation give the second lesson: one device shows an event, correlation shows the campaign. Then resilience: RPO drives backup frequency and RTO drives failover, cold, warm and hot sites trade cost against speed, and a plan that has never been tested is a guess with a cover page. If the group is strong, ask them to choose a site for a named business and justify the cost.

Slides 29-34 · Core services and remote access (23 minutes)

Recite DORA from memory and then move to the security framing: the client accepts the first offer, so a rogue server wins the race. Cover reservations, scope options and relay agents as the design vocabulary, then IPv6 addressing methods as four rows with different owners (host, server, host plus server, host rotating). DNS is the trust anchor: hierarchy, record types, time to live, split horizon and DNSSEC. Time is forensic infrastructure: drift breaks tickets, logs and timelines, so use several sources and authenticate them. Finish with VPNs and the four device-management paths, and make the point that each path is itself an attack surface.

Live demo (slide 29)

If a lab is available, run a DHCP exchange behind a mirrored port and show the Discover going to the broadcast address from no address at all. The moment candidates see the client accept the first Offer it receives, the rogue server attack stops being abstract.

Slides 35-36 · Kali verification (20 minutes)

Run the commands live: dig for DNS, cat of the dhclient lease file for DHCP, ntpq -p and timedatectl for time, ss -tln for every listener, and journalctl and auth.log for SSH evidence. Stress the analyst habit: correlate SSH and RDP session times, users and flows for a complete picture. Frame the box as an attack surface: every unexpected listener is a path in, and closing unused services is the first hardening step an analyst takes.

Live demo (slide 36)

```
grep 'Failed password' /var/log/auth.log | wc -l shows brute-force volume; ss -tunap | grep -E ':(22|53|123)' shows the service ports in use; an unexpected listener on a high port is an indicator of compromise.
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Slides 37-40 · Quiz, review and close (17 minutes)

Run the quiz, review low scores, walk the takeaways and assign the checklist. Preview Session 4, where the Security+ core begins with fundamental security concepts. Close with the one habit from the narration: before you change anything, ask what breaks if it fails, and what you will do when it does.

Common misconceptions and corrections

Misconception	Correction
"A port number tells you the service is secure"	A port tells you which service answers; port 23 is Telnet in plaintext, and port 22 is Secure Shell, and they are often confused
"SFTP and FTPS are the same thing"	SFTP is file transfer inside Secure Shell on port 22; FTPS wraps FTP in TLS, usually on port 990
"Broadcast and multicast are the same"	A broadcast asks every device on the segment; a multicast reaches only the devices that joined the group
"Documentation is administration, not security"	A wrong diagram sends response teams to the wrong rack; documentation is what makes an incident recoverable
"End of Life and End of Support mean the same"	End of Life means the vendor has stopped selling it; End of Support means no further patches, which is the security cliff
"Factory reset destroys the data on a switch"	Configurations, certificates and stored credentials survive a soft reset, so sanitise, then destroy or overwrite the media
"Change control is bureaucracy"	Uncontrolled change is a leading cause of outage; the change request is paperwork, but the rollback plan is the valuable part
"SNMP community strings are passwords, so they are safe"	A community string is sent in plaintext in older versions; use SNMP version 3, which adds authentication and encryption
"Full packet capture is always better than flows"	Capture costs storage and holds personal data; flows show trends, floods and capacity over months at a fraction of the cost
"A hot site removes the need for testing"	A hot site reduces recovery time, but a plan that has never been tested is still unproven, so test at least once a year
"DHCP assigns permanent addresses"	Addresses are leased; a reservation pins an address to a hardware address but the client still negotiates a lease
"IPv6 hands out addresses only from a server"	Stateless autoconfiguration lets the host build its own address from the router advertisement, with no server involved
"NTP is just a convenience"	

Misconception	Correction
	Kerberos, certificate validation and forensic timelines all depend on accurate time, so drift silently breaks authentication and evidence
"Telnet is acceptable on internal networks"	Telnet sends credentials in clear text; SSH is the encrypted replacement and is available everywhere

Assessment check

- Session objectives map to chapters 1 to 6 and close the network-foundation strand.
- The HTML quiz tags every question with its topic area (3A DNS, 3B DHCP, 3C NTP, 3D SSH) for traceable revision.
- The Wayground quiz (15 questions) gives an in-class, importable assessment of the same topics.
- Ask each candidate to name one service, one attack against it, and one control, for any three of the services covered.